

E-Commerce Sales Data

Product Performance Analysis Report

TOOLS : Python

1. Executive Summary

This report presents a complete analysis of the E-Commerce Sales Data 2024–2025 dataset, covering data preparation (ETL) and Exploratory Data Analysis (EDA) with a focus on product-level performance. The dataset contains 5,000 order records across 14 fields, including order details, customer information, product categories, and financial metrics.

The analysis evaluates products across five dimensions — total sales, quantity sold, total profit, profit margin, and average discount — to identify top performers and support inventory, pricing, and promotional decisions.

Key Findings

- Headphones Accusantium is the top-performing product overall, generating the highest total sales (₹8,57,184.20) and the highest total profit (₹1,41,873.79).
- Microwave Provident sold the highest volume, with 17 units sold — the top by quantity.
- Dumbbells Doloribus achieved the best profit margin among top products, at approximately 25%.
- Products were segmented into High, Medium, and Low margin tiers; the majority (1,917 of 3,835 unique products) fall in the Low Margin tier.
- Several Yoga Mat and Accessories products carry the highest average discount rates, at 20%.

2. Data Preparation (ETL)

Before analysis, the raw dataset was cleaned and validated to ensure accuracy and consistency:

- Loaded the raw CSV file (Ecommerce_Sales_Data_2024_2025.csv) into a structured DataFrame.
- Checked and confirmed there were no duplicate records in the dataset.
- Checked for missing values across all 14 columns — none were found.
- Cleaned column names by stripping extra whitespace.
- Converted the 'Order Date' field to a proper datetime format for time-based analysis.
- Trimmed whitespace from all text fields (Customer Name, Region, City, Category, Sub-Category, Product Name, Payment Mode).
- Validated numerical fields (Quantity, Unit Price, Discount, Sales, Profit) using summary statistics.
- Exported the cleaned dataset as Ecommerce_sales_clean.csv for downstream analysis.

Dataset Overview

Metric	Value
Total Records	5,000
Total Columns	14
Duplicate Records	0
Missing Values	0
Date Range	Order Date field (2023–2025 orders present)

Numerical Field Summary

Field	Mean	Std Dev	Min	Max
Quantity	2.99	1.41	1	5 (75th pct: 4)
Unit Price (₹)	39,760.90	22,831.78	222.00	—
Discount (%)	10.05	7.08	0	20 (75th pct: 15)
Sales (₹)	1,06,733.20	85,108.21	264.10	—
Profit (₹)	15,941.75	14,897.68	19.12	—

3. Exploratory Data Analysis — Product Performance

The product performance analysis examines each product's contribution to sales, quantity sold, profitability, profit margin, and discounting behaviour, in order to identify which products drive revenue and which drive volume.

3.1 Top 10 Products by Sales

Total sales were aggregated by product name to identify the highest revenue-generating products.

```
product_sales =(
    df.groupby("Product Name")["Sales"]
      .sum()
      .sort_values(ascending=False)
)
product_sales.head(10)
```

```
top_10_product =product_sales.head(10)
plt.figure(figsize=(10,6))
sns.barplot(
    x=top_10_product.values,
    y=top_10_product.index
)
plt.title("Top 10 Product by sales ")
plt.xlabel("Total sales")
plt.ylabel("Product name")
plt.show()
```

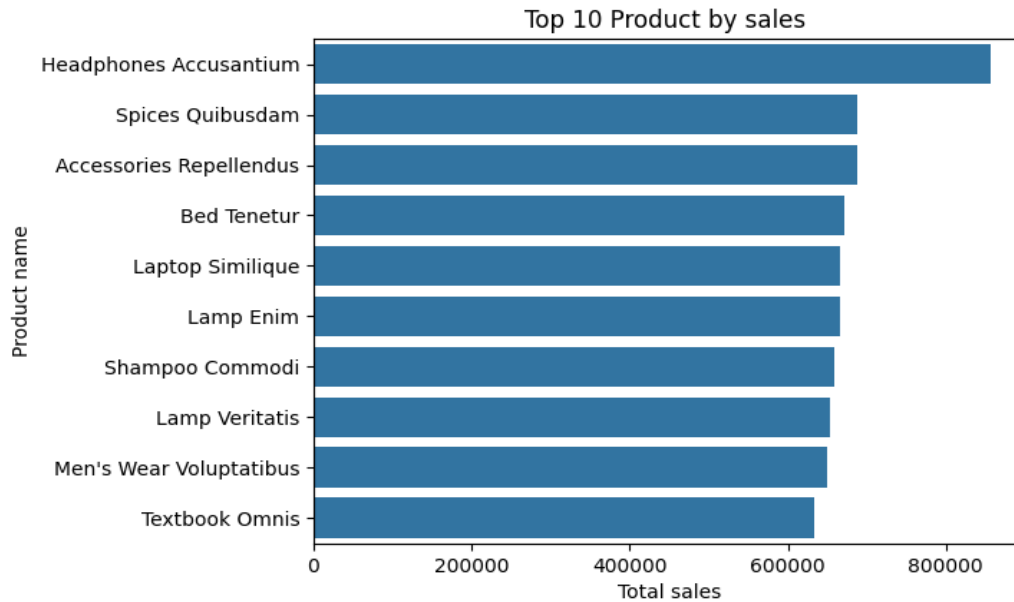


Figure 1: Top 10 Products by Total Sales

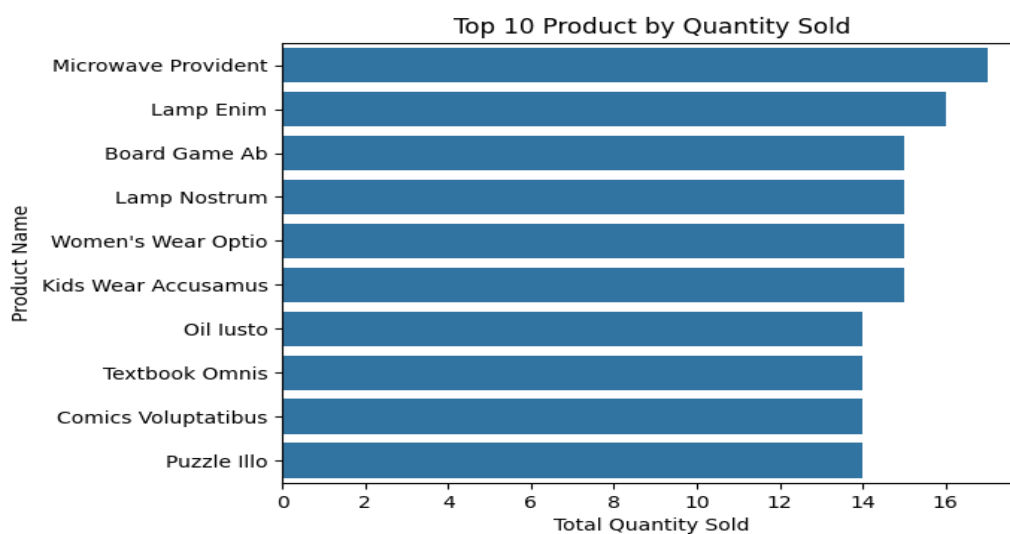
Rank	Product Name	Total Sales (₹)
1	Headphones Accusantium	8,57,184.20
2	Spices Quibusdam	6,87,651.25
3	Accessories Repellendus	6,87,083.30
4	Bed Tenetur	6,70,994.80
5	Laptop Similique	6,66,640.60
6	Lamp Enim	6,65,920.20
7	Shampoo Commodi	6,59,546.60
8	Lamp Veritatis	6,52,909.50
9	Men's Wear Voluptatibus	6,49,048.00
10	Textbook Omnis	6,33,246.25

3.2 Top 10 Products by Quantity Sold

Quantity sold was aggregated by product to identify the highest-volume products — useful for inventory and stock planning.

```
product_quantity =(
    df.groupby("Product Name")["Quantity"]
      .sum()
      .sort_values(ascending=False)
)
product_quantity.head(10)
```

```
top_10_product_quantity=product_quantity.head(10)
plt.figure(figsize=(10,6))
sns.barplot(
    x=top_10_product_quantity.values,
    y=top_10_product_quantity.index
)
plt.title("Top 10 Product by Quantity Sold")
plt.xlabel("Total Quantity Sold")
plt.ylabel("Product Name")
plt.show()
```



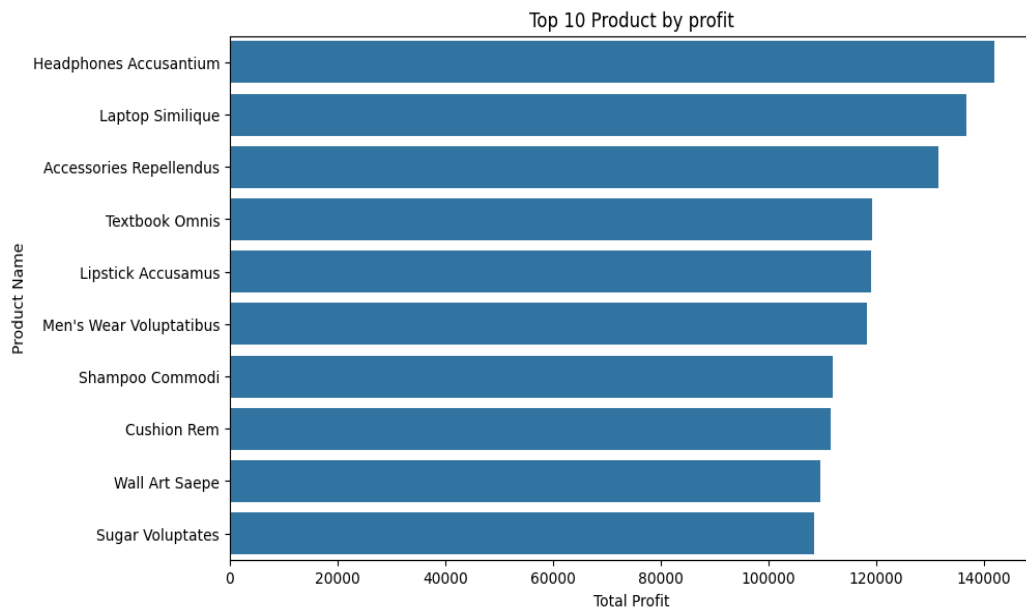
Rank	Product Name	Units Sold
1	Microwave Provident	17
2	Lamp Enim	16
3	Board Game Ab	15
4	Lamp Nostrum	15
5	Women's Wear Optio	15
6	Kids Wear Accusamus	15
7	Oil Iusto	14
8	Textbook Omnis	14
9	Comics Voluptatibus	14
10	Puzzle Illo	14

3.3 Top 10 Products by Profit

Total profit was aggregated by product to identify which products contribute the most to the bottom line.

```
product_profit=(
    df.groupby("Product Name")["Profit"]
      .sum()
      .sort_values(ascending=False)
)
product_profit.head(10)
```

```
top_10_product_profit = product_profit.head(10)
plt.figure(figsize=(10,6))
sns.barplot(
    x=top_10_product_profit.values,
    y=top_10_product_profit.index
)
plt.title("Top 10 Product by profit")
plt.xlabel("Total Profit")
plt.ylabel("Product Name")
plt.show()
```



Rank	Product Name	Total Profit (₹)
1	Headphones Accusantium	1,41,873.79
2	Laptop Similique	1,36,751.80
3	Accessories Repellendus	1,31,526.41
4	Textbook Omnis	1,19,166.66
5	Lipstick Accusamus	1,19,017.65
6	Men's Wear Voluptatibus	1,18,300.59
7	Shampoo Comodi	1,11,950.64
8	Cushion Rem	1,11,530.63
9	Wall Art Saepe	1,09,548.98
10	Sugar Voluptates	1,08,494.16

3.4 Sales vs. Profit — Combined View

Comparing sales and profit side-by-side for the top 10 products by sales highlights how efficiently revenue converts into profit for each product.

```

product_performance =(
    df.groupby('Product Name')[['Sales', 'Profit']]
    .sum()
    .sort_values("Sales" , ascending=False)
)
product_performance.head(10)

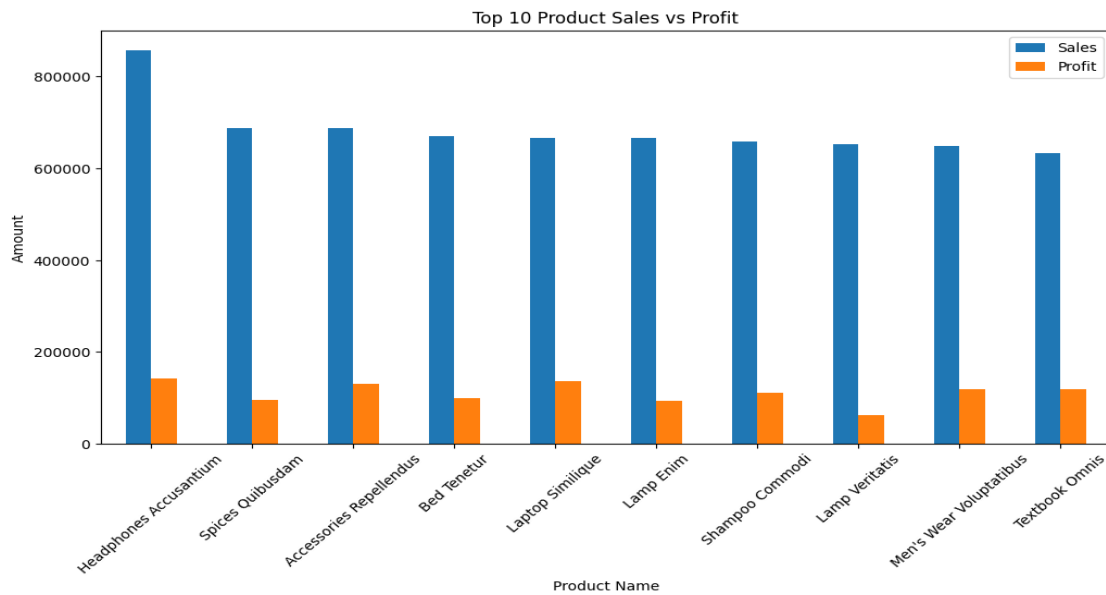
```

	Sales	Profit
Product Name		
Headphones Accusantium	857184.20	141873.79
Spices Quibusdam	687651.25	96587.33
Accessories Repellendus	687083.30	131526.41
Bed Tenetur	670994.80	100233.59
Laptop Similique	666640.60	136751.80
Lamp Enim	665920.20	93638.94
Shampoo Commodi	659546.60	111950.64
Lamp Veritatis	652909.50	62922.96
Men's Wear Voluptatibus	649048.00	118300.59
Textbook Omnis	633246.25	119166.66

```

top_product_performance =product_performance.head(10)
top_product_performance.plot(
    kind="bar",
    figsize=(12,6)
)
plt.title("Top 10 Product Sales vs Profit")
plt.xlabel("Product Name")
plt.ylabel("Amount")
plt.xticks(rotation = 45)
plt.legend(["Sales","Profit"])
plt.show()

```



3.5 Profit Margin Analysis

Profit Margin (%) was calculated as $(\text{Total Profit} \div \text{Total Sales}) \times 100$ for each product, identifying which products are most efficient rather than simply highest-earning.

```
product_margin = (  
    df.groupby('Product Name')[['Sales', 'Profit']]  
    .sum()  
)  
  
product_margin['Profit Margin %'] = (  
    product_margin['Profit'] /  
    product_margin['Sales']  
) * 100  
  
product_margin = product_margin.sort_values(  
    'Profit Margin %',  
    ascending=False  
)  
  
product_margin.head(10)
```

```

top_10_margin = product_margin.head(10)

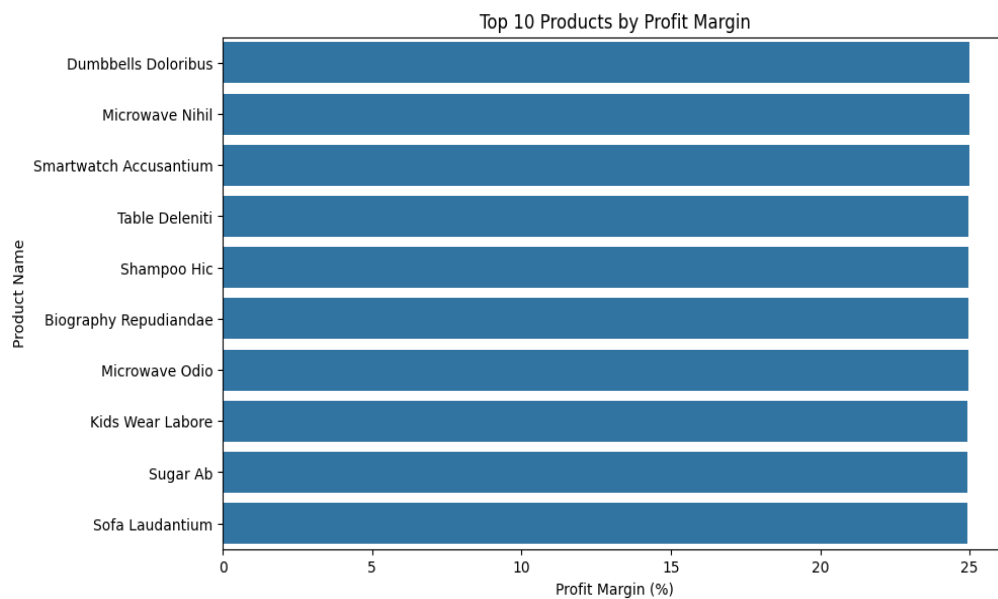
plt.figure(figsize=(10, 6))

sns.barplot(
    x=top_10_margin['Profit Margin %'],
    y=top_10_margin.index
)

plt.title('Top 10 Products by Profit Margin')
plt.xlabel('Profit Margin (%)')
plt.ylabel('Product Name')

plt.show()

```



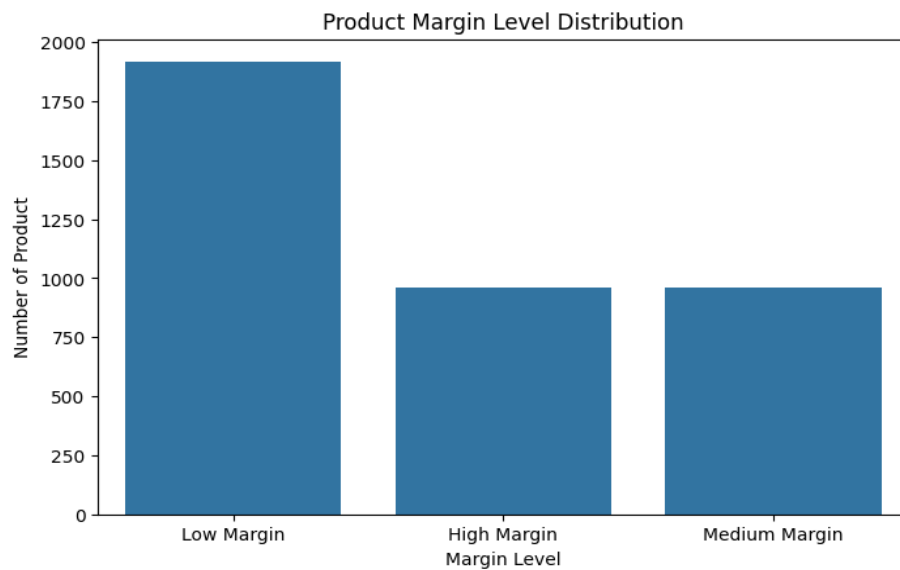
Rank	Product Name	Profit Margin (%)
1	Dumbbells Doloribus	24.996%
2	Microwave Nihil	24.979%
3	Smartwatch Accusantium	24.978%
4	Table Deleniti	24.974%
5	Shampoo Hic	24.959%
6	Biography Repudiandae	24.948%
7	Microwave Odio	24.947%
8	Kids Wear Labore	~24.9%

3.6 Profit Margin Segmentation

Products were segmented into three margin tiers using the 25th and 75th percentile margin values as cut-offs, via NumPy's `np.select`: High Margin (≥ 75 th percentile), Medium Margin (25th–75th percentile), and Low Margin (below 25th percentile).

```
margin_75 =product_margin["Profit Margin %"].quantile(0.75)
margin_25 =product_margin["Profit Margin %"].quantile(0.25)
product_margin["Margin Level"] =np.select(
    [
        product_margin["Profit Margin %"]>=margin_75,
        product_margin["Profit Margin %"] >=margin_25
    ],
    [
        "High Margin",
        "Low Margin"
    ],
    default="Medium Margin"
)
```

```
margin_distribution =product_margin["Margin Level"].value_counts()
plt.figure(figsize=(8,5))
sns.barplot(
    x=margin_distribution.index,
    y=margin_distribution.values
)
plt.title("Product Margin Level Distribution")
plt.xlabel("Margin Level")
plt.ylabel("Number of Product")
plt.show()
```



Margin Level	Number of Products
Low Margin	1,917
High Margin	959
Medium Margin	959

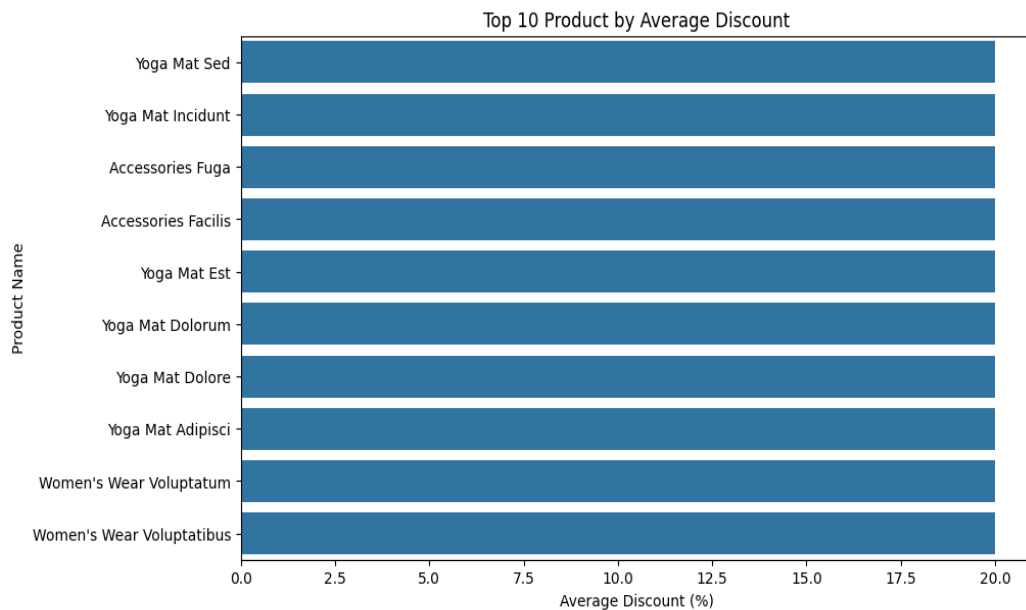
3.7 Average Discount Analysis

The average discount percentage was calculated per product to identify which products are discounted most heavily, which can help evaluate promotional strategy and its impact on margins.

```
product_discount =(
    df.groupby("Product Name")["Discount"]
    .mean()
    .sort_values(ascending=False)
)
product_discount.head(10)
```

```
Product Name
Yoga Mat Sed          20.0
Yoga Mat Incidunt     20.0
Accessories Fuga      20.0
Accessories Facilis   20.0
Yoga Mat Est          20.0
Yoga Mat Dolorum      20.0
Yoga Mat Dolore       20.0
Yoga Mat Adipisci     20.0
Women's Wear Voluptatum 20.0
Women's Wear Voluptatibus 20.0
Name: Discount, dtype: float64
```

```
top_10_discount =product_discount.head(10)
plt.figure(figsize=(10,6))
sns.barplot(
    x=top_10_discount.values,
    y=top_10_discount.index
)
plt.title("Top 10 Product by Average Discount")
plt.xlabel("Average Discount (%)")
plt.ylabel("Product Name")
plt.show()
```



The top discounted products (Yoga Mat variants, Accessories, and Women's Wear items) each carry an average discount of 20%, the maximum discount level observed in the dataset.

4. Conclusion & Recommendations

The product performance analysis provides a clear, multi-dimensional view of the E-Commerce Sales Data 2024–2025 dataset:

- Headphones Accusantium is the standout product — leading in both total sales and total profit — and should be prioritized for stock availability and marketing focus.
- Microwave Provident leads in unit volume, indicating strong demand that should be reflected in inventory planning.
- High-margin products such as Dumbbells Doloribus, Microwave Nihil, and Smartwatch Accusantium, though not top revenue earners, are highly efficient and worth promoting for profitability.
- With 1,917 products (roughly half) falling in the Low Margin tier, there is an opportunity to review pricing or cost structures for a large share of the catalog.
- Heavily discounted categories (Yoga Mats, Accessories, Women's Wear) should be monitored to ensure discounting is not eroding profit margins disproportionately.

Overall, this analysis supports data-driven decisions across inventory planning, pricing strategy, and promotional targeting by identifying which products drive revenue, which drive volume, and which are most profitable per rupee of sale.